



Department of Computer Science and Engineering

Jessore University of Science and Technology

Semester Final Examination-2022

B.Sc. (Engg.) 1st year 2nd Semester, Session: 2020-21

Course Code: CSE 1203

Time: 3.00 hours

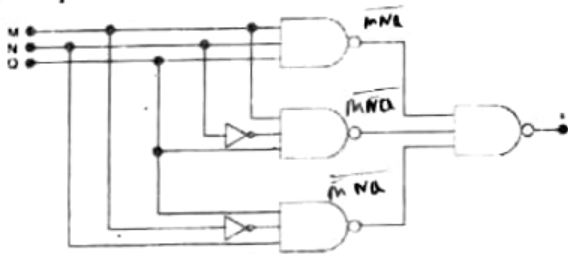
Course Title: Digital Logic Design

Marks: 72

[N.B. Answer any 6 set questions from following 8 set questions. The figures in the margin indicate full marks.]

1. a) Simplify these expressions into simplified SOP form: 12
 - a) $W = AA + AB' + A'BB'C$
 - b) $Y = (A+B'+C'+D')' + A'B + CD + AC'E' + AC'$
2. a) Draw the truth table of the following Boolean Function. Then draw the Karnaugh Map to find the simplified Boolean Expression. Finally draw the logic diagram of – 12

$$F(A,B,C,D) = \sum(0,1,2,6,9,12) + \sum d(3,4,8,13)$$
3. a) Number system is a writing system of expressing numbers. To design efficient digital systems an engineer must have the knowledge of number system conversion. Also, s/he must be able to represent negative binary numbers using sign and magnitude, 1's complement and 2's complement methods. Now, Answer the following questions. 06
 - I. Convert the number to its equivalent form: $(242.625)_{10} = (?)_2$.
 - II. Convert the number to its equivalent form: $(AD0D)_{16} = (?)_8$.
 - III. Perform the arithmetic operation using 1's complement:
 - iv. $(-87)_{10} + (-23)_{10} = (?)_{10}$

Assume that, the representation is an 8-bit register.
- b) Determine the value of 'x' if $(211)_x = (152)_8$. 04
- c) Represent decimal number 8620 in i) BCD and ii) excess-3 code. 02
4. a) Design a 2-to-4 Binary Decoder with proper diagram, truth table and circuit diagram. 6
 - b) The decoder is an electronic device that is used to convert a digital signal to an analog signal. It allows a single input line and produces multiple output lines. The decoders are used in many communication projects that are used to communicate between two devices. How a decoder can be used for Automatic Wireless Health Monitoring System in Hospitals for Patients 6
5. a) Design a 4:1 Multiplexer. Also describe Advantages and Disadvantages of Multiplexer. 12
6. a)  05
 - I. Simplify the circuit of figure using Boolean algebra.
 - II. Change each gate in Problem to a NOR gate, and simplify the circuit using Boolean algebra.
- b) Construct a 16×1 mux using five 4×1 mux only. 04
- Prove that $f1 = m0 + m2 + m5 + m7 = M1.M3.M4.M6$. 03
- a) Differentiate between Latch and Flip Flop. 6
- b) Describe D-Flip Flop. Also show the advantages of using D-Flip Flop instead of S-R Flip Flop 6
8. a) What is shift register? Show the structure of Parallel IN Serial OUT shift register. 6
- b) What are the features of the Ripple Counter. 6